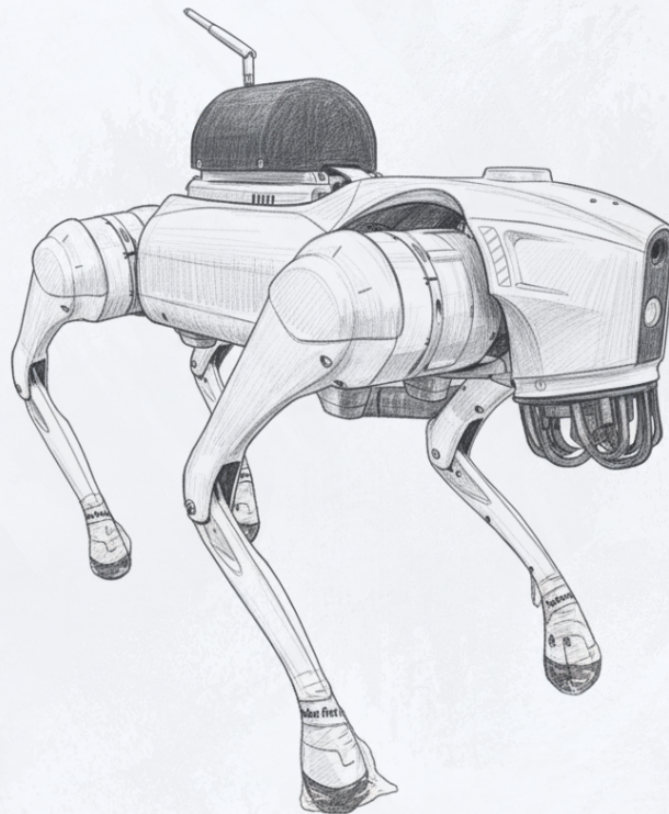


QUADRUPED ROBOTICS

QRE GO2 / Steamdeck USER MANUAL



MYBOT
SHOP

-  info@mybotshop.de
-  support@mybotshop.de
-  forum.mybotshop.de
-  docs.mybotshop.de



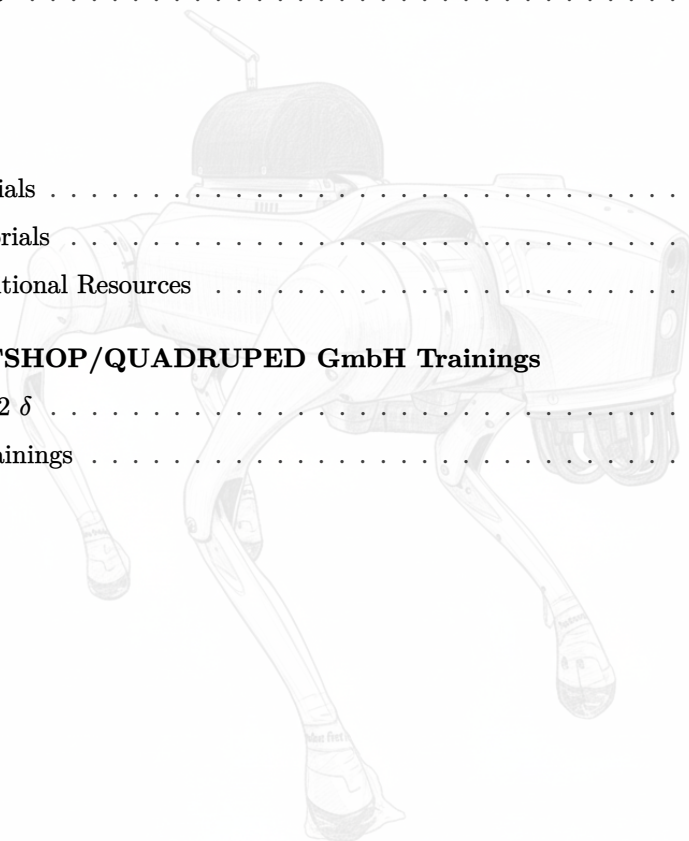
QUADRUPED
ROBOTICS

-  info@quadruped.de
-  support@quadruped.de
-  forum.mybotshop.de
-  docs.quadruped.de

Contents

1	Attention	3
2	Disclaimer	4
3	Quick-Start	5
3.1	Powering-on	5
3.2	Webserver	5
3.3	Robot Interface	8
3.4	Visualization	11
3.5	Tele-operation	11
3.6	Action Modes	11
3.7	Low-Level Control	12
4	Navigation	14
4.1	SLAM (Simultaneous Localization and Mapping)	14
4.2	Odometric Navigation	14
4.3	Map Navigation	15
5	Core Packages	16
5.1	Auto Drivers Startup	16
6	Auxiliary Sensors	17
6.1	Depth-Camera - Intel Realsense D435i	17
7	Steamdeck	17
7.1	Steamdeck - Basic Controls	17
7.2	Steamdeck Activation	17
7.3	Steamdeck - Termination	19
7.4	Steamdeck - Robot Controls	19
7.5	Steamdeck - Rviz2 Controls	19
8	Installation	20
8.1	NVIDIA Orin NX Developer Kit (Inbuilt GO2 EDU)	20
8.2	Host PC	22
9	Robotic Manipulator Safety Guidelines	23
9.1	Work Area Safety	23
9.2	Electrical Safety	23
9.3	Manipulation Safety	23
9.4	Emergency Response	23

9.5	Data Security and Privacy	24
9.6	Human Interaction Safety	24
9.7	Residual Risks	24
10	Autonomous Mobile Robot Safety Guidelines	25
10.1	Work Area Safety	25
10.2	Electrical Safety	25
10.3	Navigation Safety	25
10.4	Emergency Response	25
10.5	Data Security and Privacy	26
10.6	Human Interaction Safety	26
10.7	Residual Risks	26
11	Support	27
12	Tutorials	28
12.1	Ubuntu Tutorials	28
12.2	ROS2 — Tutorials	28
12.3	ROS2 — Additional Resources	28
13	ROS — MYBOTSHOP/QUADRUPED GmbH Trainings	30
13.1	ROS1 to ROS2 δ	30
13.2	Additional Trainings	32

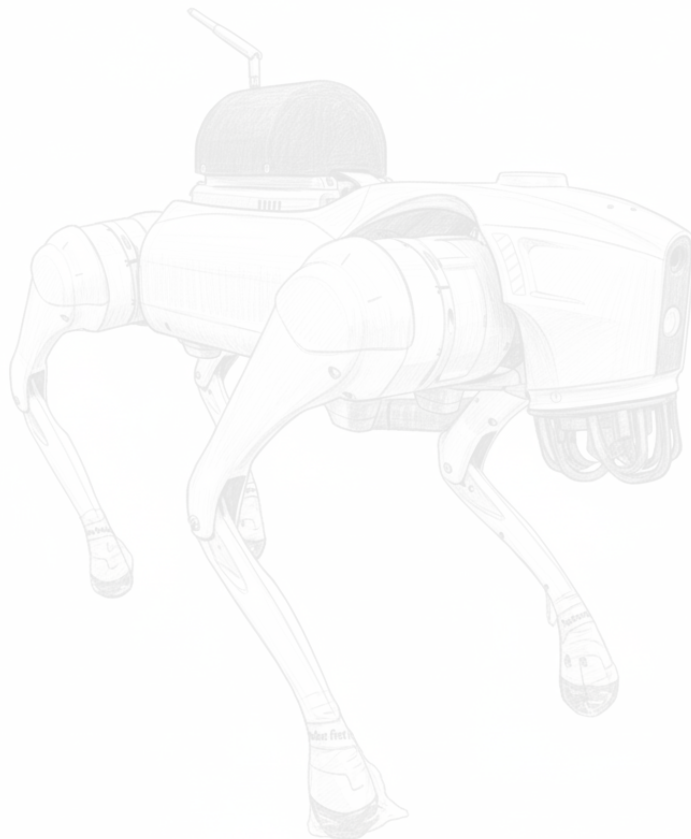


1 | Attention

Read this document carefully. In case of doubt, it is essential to consult specialists, experts, or manufacturers of the assemblies used! The robot must not be put into operation before clarification.

In case of doubt, please consult the guides below or contact a specialist, expert, or manufacturer of the assemblies used! The robot must not be put into operation before clarification. Contact information is available in the forums of the Guideline Section [11](#).

Do not allow persons who are not familiar with the robotics or these instructions to operate the robot. Robots are dangerous in the hands of untrained users.



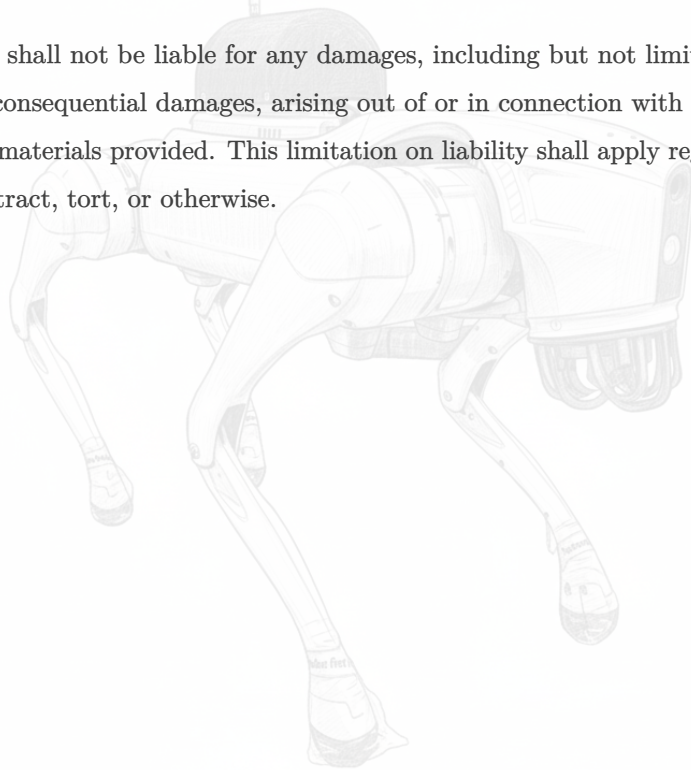
2 | Disclaimer

The provided robot, sold by *MYBOTSHOP GmbH*, is an R&D device and does not have CE Marking and/or Certificate of Incorporation. Basic ROS understanding is required. If you are not familiar with ROS, we recommend checking the [ROS2 Wiki](#) first.

Please note that the quoted robot is considered a partly completed machine under the *Machinery Directive 2006/42/EC*, and it does not carry a CE marking.

The client acknowledges and agrees that any information or materials provided by *MYBOTSHOP GmbH* are for R&D purposes only. Any services are provided "AS IS" and without any representation or warranty of any kind, express or implied, including but not limited to any warranty of merchantability, fitness for a particular purpose, non-infringement, or any other warranty.

MYBOTSHOP GmbH shall not be liable for any damages, including but not limited to direct, indirect, special, incidental, or consequential damages, arising out of or in connection with the use or inability to use the information or materials provided. This limitation on liability shall apply regardless of the form of action, whether in contract, tort, or otherwise.



3 | Quick-Start

3.1 | Powering-on



- Power on the robot by pressing the battery once and then again for 3 seconds.
- Ensure the battery is fully charged before powering on the robot.
- If a unitree remote is available, you can also power on the remote by pressing the power once and then for 3 seconds.

3.2 | Webserver

This module should come pre-installed for heavy integration projects. It should be accessible directly at <http://192.168.123.18:9000/> or the WiFi ip to which the robot is connected. The GO2 webserver can be configured via the config file in `go2_webserver` ros2 package located in `/opt/mybotshop/src/mybotshop/go2_webserver/config/robot_webserver.yaml`

3.2.1 | Login

- Username: **admin**
- Password: **mybotshop**

3.2.2 | Dashboard

- View IP Address of the GO2
- View System load

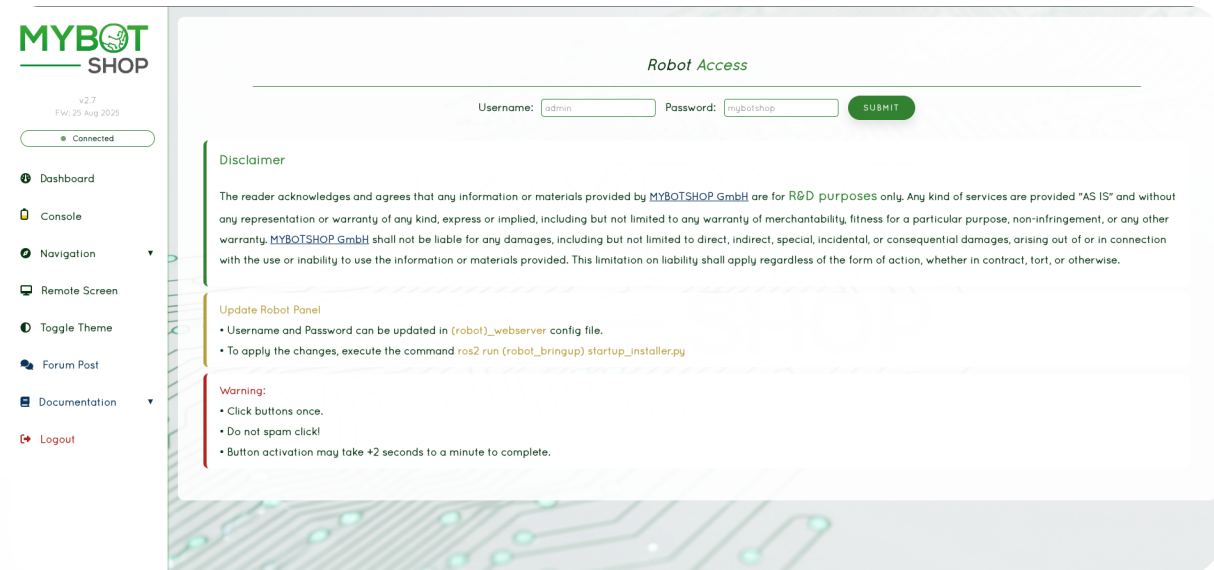


Figure 3.1: Web Login

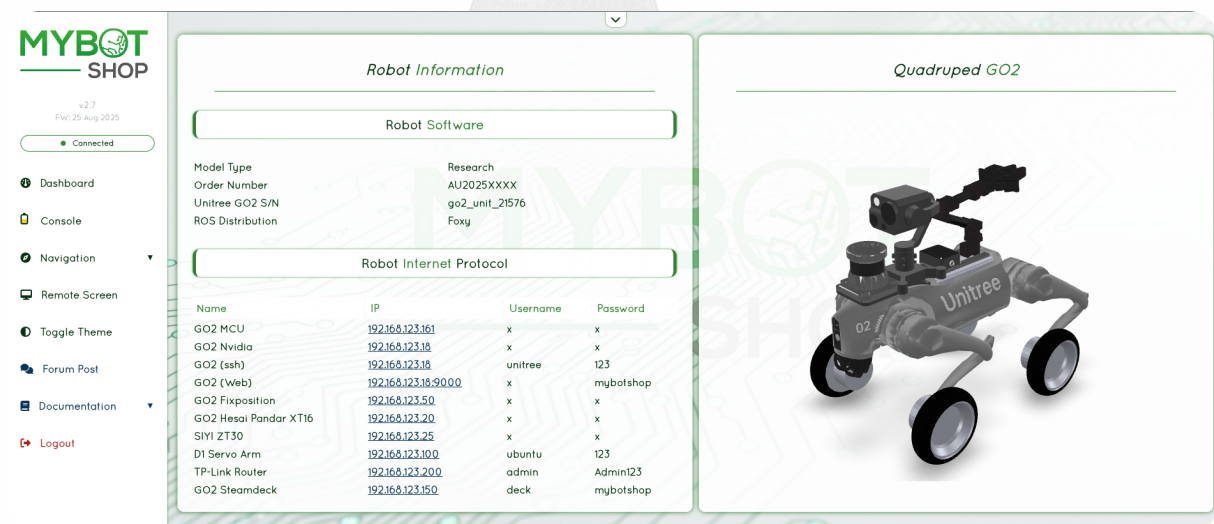


Figure 3.2: Web Dashboard

3.2.3 | Console

- Enable the GO2 ROS2 Services
- Disable the GO2 ROS2 Services
- Record System GO2 logs
- Battery status
- Pre-configured action buttons
 - Switch Gaits
 - Stand up
 - Sit down

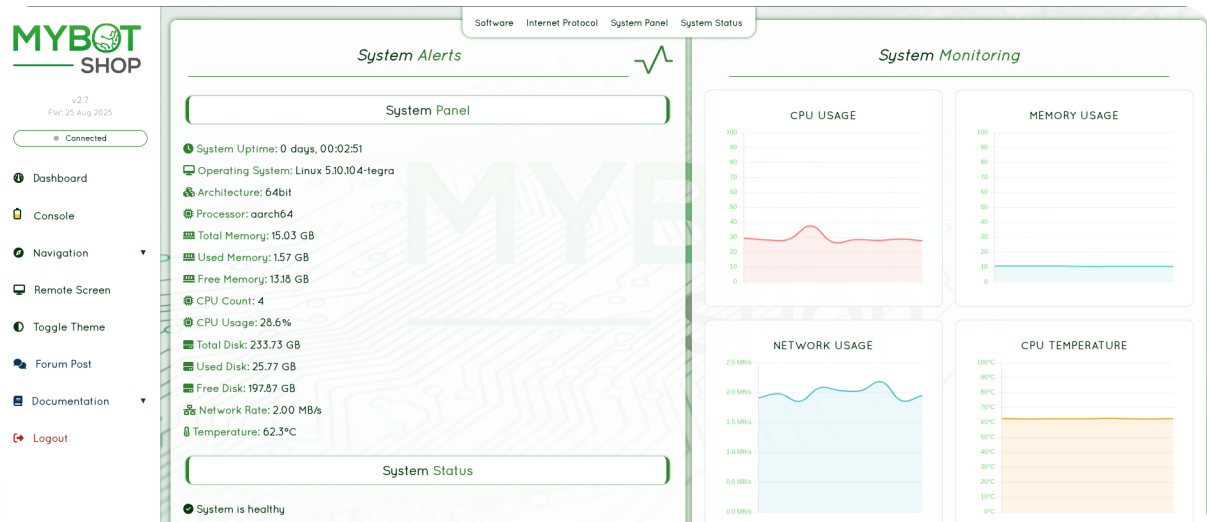


Figure 3.3: Web System

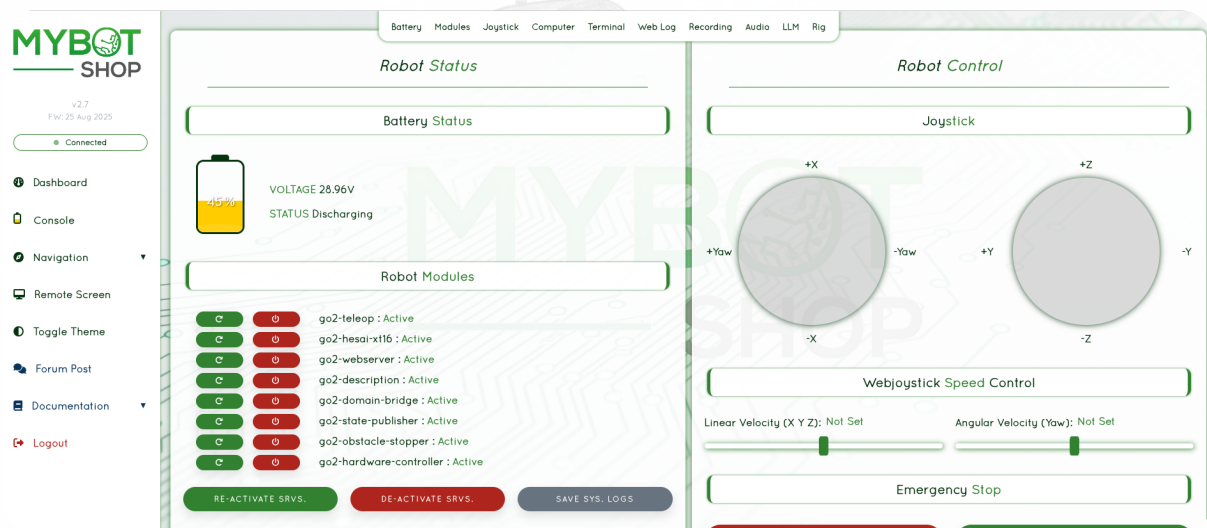


Figure 3.4: Web Console

□ Adapatable to new ros2 services

■ Web Joystick

■ Online image stream

■ STT to A.I to TTS (Disabled)

3.2.4 | Remote Desktop

■ On-board screen of the Unitree GO2's computer

3.2.5 | Navi Indoor

■ Disabled

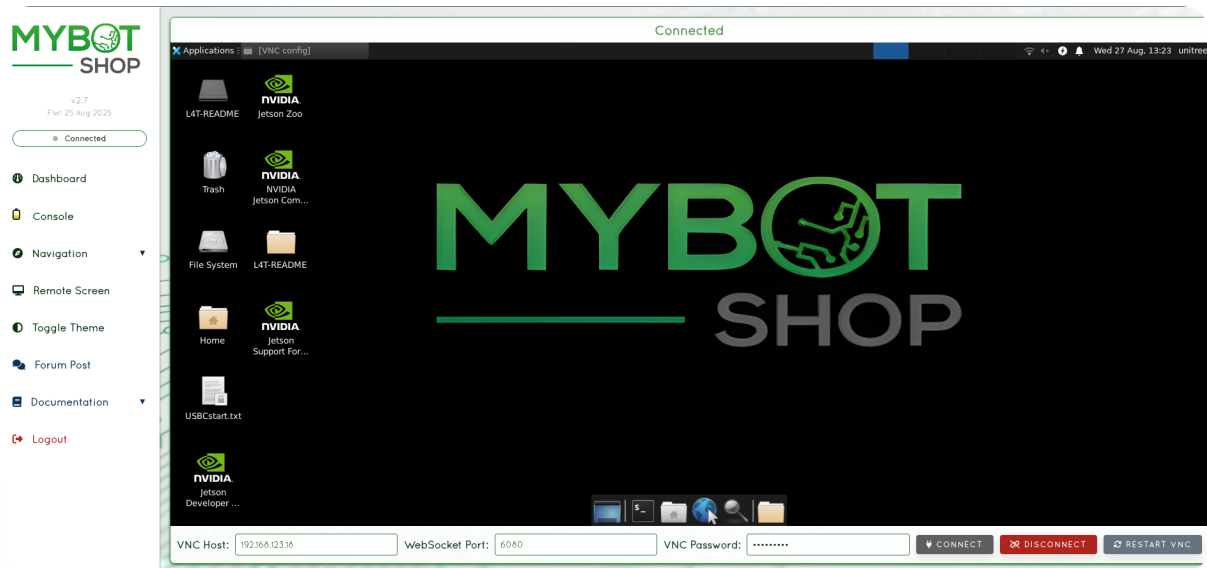


Figure 3.5: Web Remote Desktop

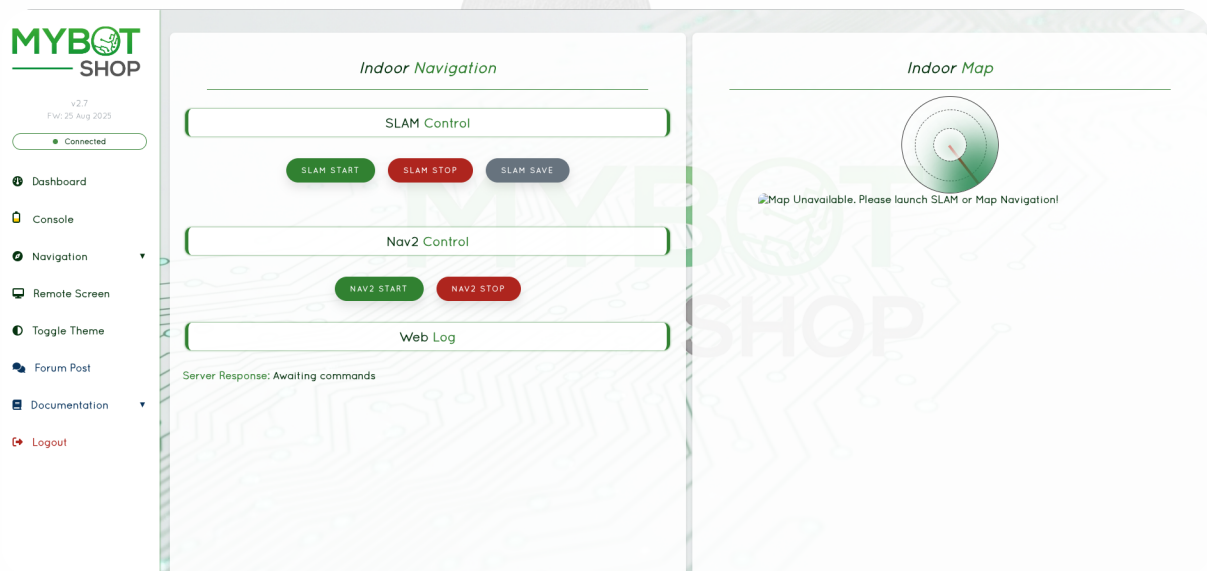


Figure 3.6: Web Navigation Indoor

3.2.6 | Navi Outdoor

■ Disabled

3.2.7 | LLM Interface

■ Disabled

3.3 | Robot Interface

Instructions for interfacing with the robot using **Ubuntu 20.04** and **ROS2 Foxy**.

Note: This procedure should be followed after setting up and pairing with the GO2 Edu. Furthermore, all of the GO2's functionality should be verified via the app. Instructions for set up can be found at [GO2 Docs](#) also available in section 11. This guide builds upon the information from the docs. The QRE GO2 operates in ROS DOMAIN ID 10. Ensure that your ROS2 environment is set to the same domain ID when interfacing with the GO2's QRE packages.



Figure 3.7: The Ethernet port at the back of the GO2 can be utilized for setting up communication via LAN.

3.3.1 | Static Network Connection

For the first time, one needs to connect through a **LAN** cable to configure the robot's network.

To create a static connection in your PC (not the robots), in Ubuntu go to Settings → **Network** then click on + and create a new connection.

1. Change the connection to **Manual** in the **IPv4** settings.
2. Set the **Address IP** as **192.168.123.51** and the **Netmask** as **24**.
3. Click save and restart your network.

After a successful connection, check the host's local IP by typing in the Host PC's terminal:

```
ifconfig
```

Now, ping the robot:

```
ping 192.168.123.18
```

Access the robot via SSH:

```
ssh -X unitree@192.168.123.18
```

The default password is:

```
123
```

3.3.2 | GO2 IP Addresses

Robot	IP Addresses	Username	Password
GO2 MCU	192.168.123.161	-	-
GO2 External	192.168.123.18	unitree	123
GO2 External Web	http://192.168.123.18:9000	admin	mybotshop
Unitree D1 Arm	192.168.123.100	ubuntu	123
ZT30	192.168.123.25	-	-
Ouster	192.168.123.20	-	-
Livox Mid360	192.168.123.20	-	-
Router SSID	192.168.123.200	GO2XXXXXX-5G	mybotshop
Router Web	192.168.123.200	-	Admin123

Note: Sometimes other networks can cause disruptions when connecting to the GO2. It is best to have only your connection to the robot active and all others inactive.

3.3.3 | Interfacing

- Firstly, connect to the GO2 as described in the Network section 3.3.
- Secondly, open several ssh sessions into the go2 via

```
ssh -X unitree@192.168.123.18
```

- Verify that the robot is processing information by typing the command in an ssh session.

```
ros2 topic echo /sportmodestate
```

- If data is displayed, the drivers are running.

3.4 | Visualization

You can view the GO2's current state by typing in one of the ssh sessions:

```
ros2 launch go2_viz view_robot.launch.py
```

3.5 | Tele-operation

You can teleoperate the GO2 by running the following in one of the ssh sessions, be sure to change the id to the robots id:

```
ROS_DOMAIN_ID=10 ros2 run teleop_twist_keyboard teleop_twist_keyboard  
--ros-args --remap cmd_vel:=/go2_unit_001/cmd_vel
```

3.6 | Action Modes

Available GO2 Modes via ROS2 Services:

```
- balance_stand  
- recovery_stand  
- stand_down  
- stand_up  
- stretch  
- damp  
- sit  
- rise_sit  
- front_jump  
- front_pounce  
- front_flip  
- stop_move
```

Available GO2 Functionalities via ROS2 Services:

```
- Light level: 0-10  
- Volume level: 0-10
```

Note: Pressing the tab button on the keyboard will autocomplete the service request.

■ Example of standing up

```
ROS_DOMAIN_ID=10 ros2 service call /go2_unit_001/modes \  
go2_interface/srv/Go2Modes "{request_data: 'stand_up'}"
```

- Example of standing down

```
ROS_DOMAIN_ID=10 ros2 service call /go2_unit_001/modes \
go2_interface/srv/Go2Modes "{request_data: 'stand_down'}"
```

- Example of changing light level 0-10

```
ROS_DOMAIN_ID=10 ros2 service call /go2_unit_001/light \
go2_interface/srv/Go2Light "{light_level: 10}"
```

- Example of changing volume level 0-10

```
ROS_DOMAIN_ID=10 ros2 service call /go2_unit_001/volume \
go2_interface/srv/Go2Volume "{volume_level: 4}"
```

Note: At times the mode does not activate correctly such as stand, in those cases use the [recovery stand](#).

3.7 | Low-Level Control

The low-level control for GO2 can be directly used via Unitree's provided example in their [documentation](#). Running the provided **qre_go2** driver with the low-level commands from Unitree examples should both work concurrently.

An example to test the low-level control is to suspend the go2.

- Verify if the go2_bringup is running via:

```
sudo service go2-platform status
```

- Lie down the robot

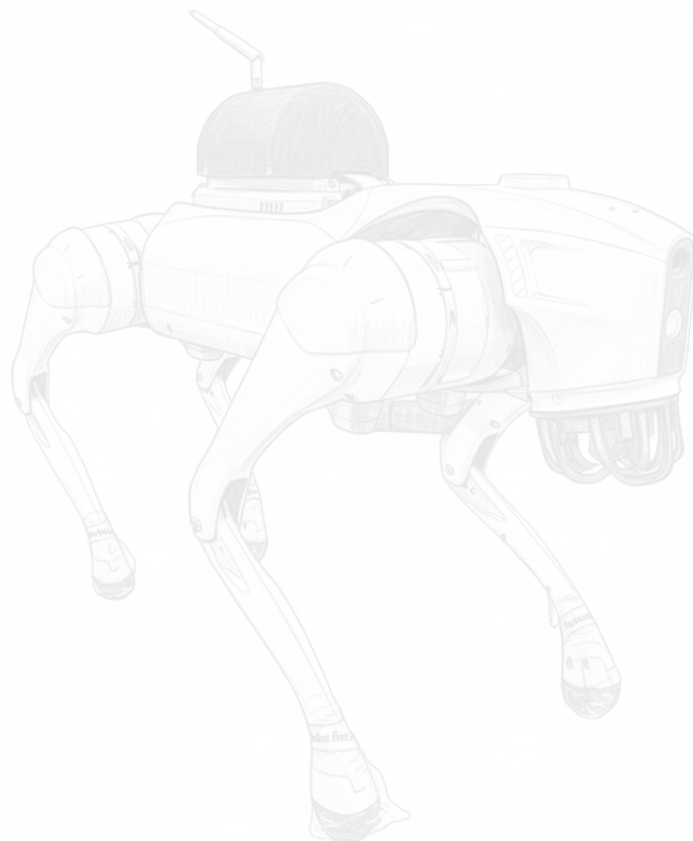
```
ROS_DOMAIN_ID=10 ros2 service call /go2_unit_001/modes \
go2_interface/srv/Go2Modes "{request_data: 'stand_down'}"
```

- Put motors in damp state

```
ROS_DOMAIN_ID=10 ros2 service call /go2_unit_001/modes \
go2_interface/srv/Go2Modes "{request_data: 'damp'}"
```

- Run low-level unitree example

Note: The GO2 low-level mode should only be used if you are familiar with low-level control, and gait planning and wish to create a new/custom controller for the GO2. The GO2 in low-level mode can easily be damaged if used incorrectly.



4 | Navigation

4.1 | SLAM (Simultaneous Localization and Mapping)

To create a map for map-based navigation, the first step is to create a map of the environment.

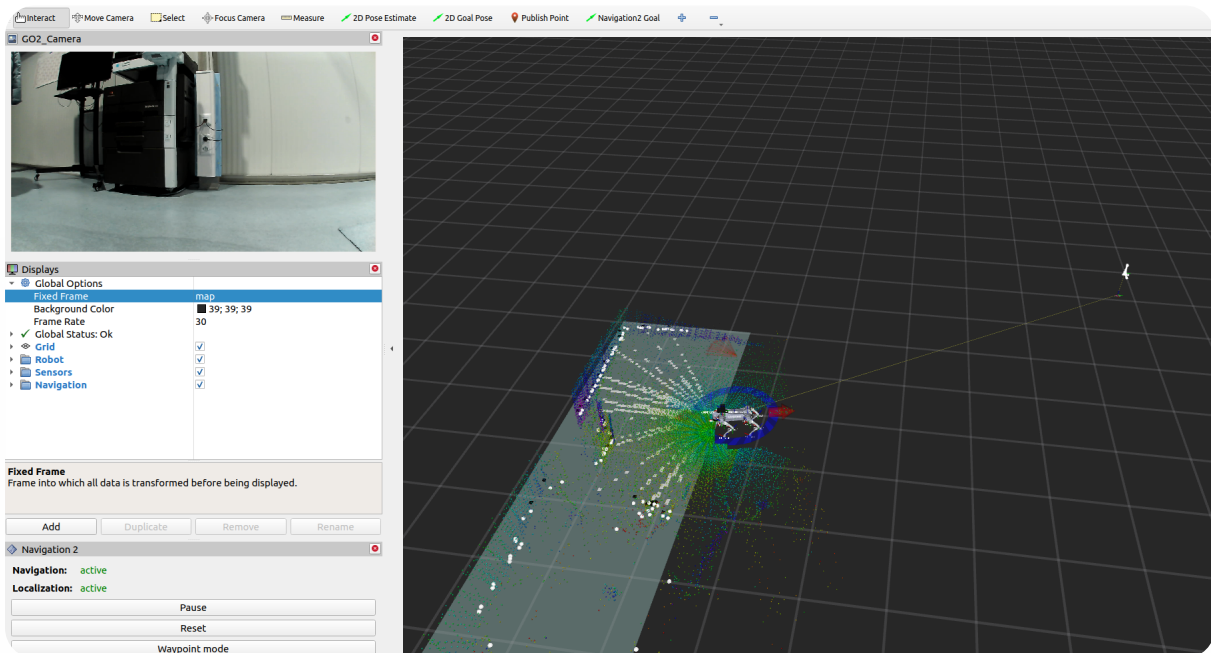


Figure 4.1: Move the robot slowly when building the map

■ Launch

```
ros2 launch go2_navigation slam.launch.py
```

- You can begin mapping using the *teleop* at **0.2m/s** with the keyboard and/or the provided Logitech controller. Once you are satisfied with your map you can export it by running the following command:

```
ROS_DOMAIN_ID=10 ros2 run nav2_map_server map_saver_cli\  
-f /opt/mybotshop/src/mybotshop/go2_navigation/maps/custom_map
```

- Rebuild so that the maps can be found (This is required if the map name is not **map_** otherwise it will directly work)

```
cd /opt/mybotshop/ && colcon build --symlink-install  
source /opt/mybotshop/install/setup.bash
```

4.2 | Odometric Navigation

To navigate without creating a map, you can simply launch the odom navi and give goals in RVIZ.

```
ros2 launch go2_navigation odom_navi.launch.py
```

4.3 Map Navigation

This will automatically take in the map created and save from the SLAM being **map**.. If the map has been named something other, then in **go2_navigation**, the parameters have to be updated.

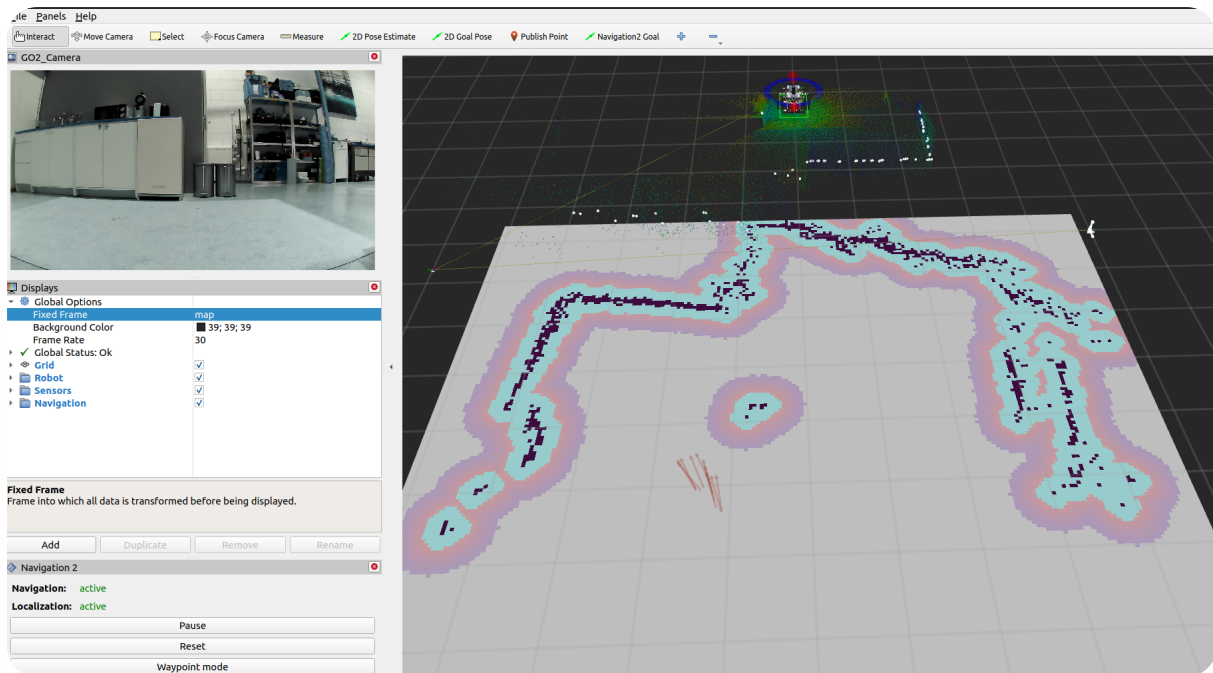


Figure 4.2: Select the 2D Pose Estimate and click and drag to where the robot corresponds in the map

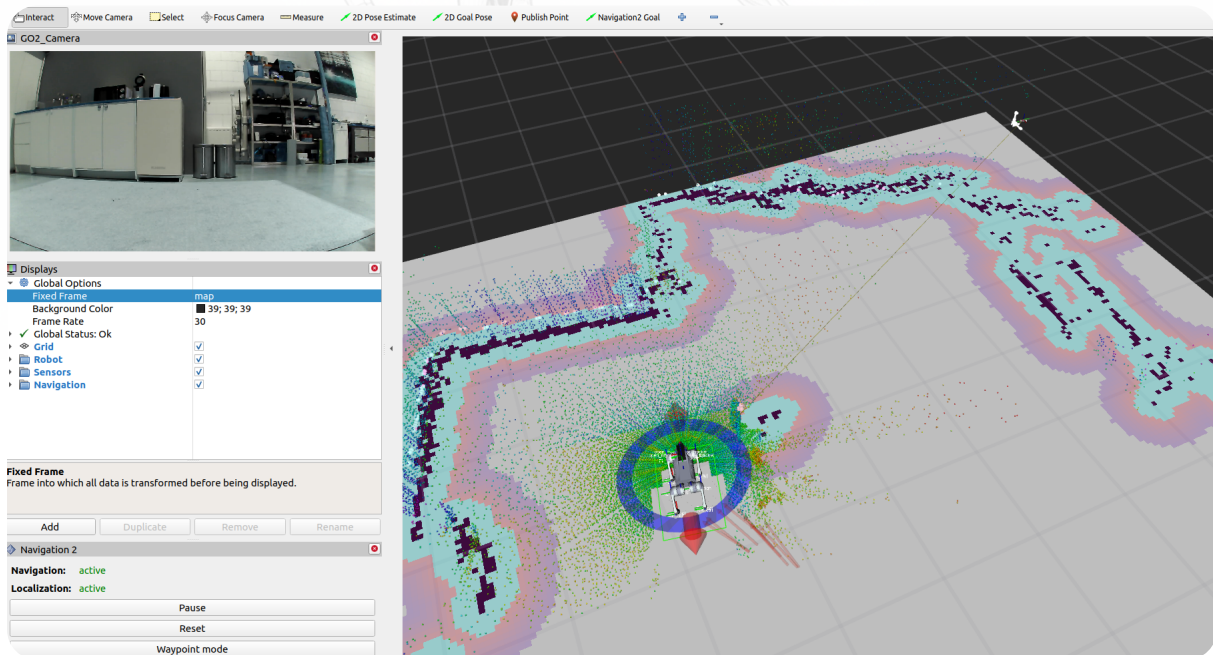


Figure 4.3: Once the robot is in the estimated location, it should self correct when the robot is moving as shown

- Ensure the map is generated and available in the ros package (i.e. after the map is saved, you have placed and performed **colcon build**)

```
ros2 launch go2_navigation map_navi.launch.py
```

5 | Core Packages

5.1 | Auto Drivers Startup

Note: The robot system can be verified by typing in the terminal. Generally, all drivers are off except the webserver and can be activated via web server or by the command:

```
sudo service go2-platform start
```

The list of all drivers can be found in the go2_bringup startup_installer.py

```
sudo service go2-webserver status
```

```
sudo service go2-platform status
```

If it is available and green, then do not launch the **go2_bringup** as it is running in the background.

Launch the go2 ros driver the communicates and publishes the state of the robot joints.

```
ros2 launch go2_bringup system.launch.py
```

The GO2 ordinarily does not have a startup job unless otherwise specified (In this case the GO2 has a startup job). The GO2 launches the **go2_bringup system.launch.py**. To verify if the startup job is available in GO2. Run the command:

```
sudo service go2-platform status
```

Note: If an error such as **Unit ros2.service could not be found.**, then it means that there is no startup installed.

1. The red marker in the service indicates that the startup job has failed.
2. Green marker indicates everything is working correctly.
3. Grey marker indicates that the service has not started yet.

In case of red or grey marker, you may restart the service via:

```
sudo service go2-platform restart
```

If you want to modify the upstart job or add other ROS launch files to it then it is recommended to add your changes to the main upstart file i.e. **system.launch.py** located in the **go2_bringup** package. Once done, save the file and run the following command to update startup job.

```
ros2 run go2_bringup startup_installer.py
```

6 | Auxiliary Sensors

6.1 | Depth-Camera - Intel Realsense D435i

- To launch the Realsense d435i, launch:

```
ros2 launch go2_depth_camera realsense_d435i.launch.py
```

- By default the Realsense D435i is off.
- The launch file is configured to enable continuous depth stream information from the Realsense d435i without lag. To further change parameters, simply change the configuration in the

```
go2_depth_camera/launch/realsense_d435i.launch.py.
```

7 | Steamdeck

7.1 | Steamdeck - Basic Controls

1. **Power button** is on the top of the Steamdeck with the power symbol.
2. **R2** for mouse left-click
3. **L2** for mouse right-click
4. **Left trackpad** for mouse scrolling
5. **Right trackpad** for mouse movement
6. **L2+R2** for middle mouse button click
7. **- - -** (The 3 dashes is located on the top right of the Steamdeck next to the Y button) Hold for 3 seconds to *gamepadmode* and hold for 2 seconds again to go to *desktop mode*¹

7.2 | Steamdeck Activation

¹Switching to desktop mode activates the cursor for the trackpad as well

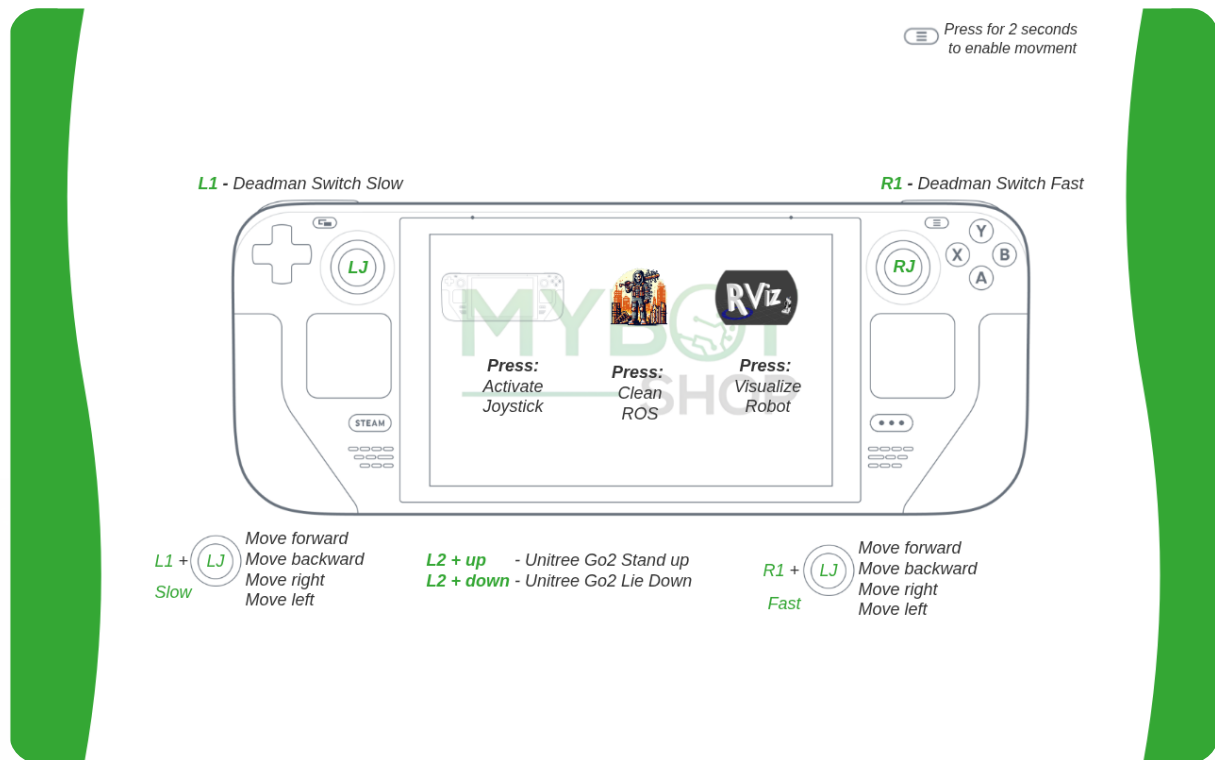


Figure 7.1: Steamdeck Shortcuts

Important: When launching the **Joylauncher**, do not touch any of the buttons until the steamdeck displays the **robot is connected** or **not connected**.

1. Ensure the Steamdeck is connected to the robot's WiFi access point. Wait 20 seconds after powering up.
 - This is automatically done!
2. Ensure the robot is powered on and operational.
3. Single tap the **Joystick Launcher** on the screen to activate the controller for the robot.
 - Do not open multiple instances of the joystick launchers, as this can cause zombie nodes to spawn.
 - Hold the - - - for 3 seconds until you sense vibration from the controller and then hold - - - for 3 seconds again. This is required to re-enable the mouse in the steamdeck.
4. Robot stand up
 - Activate all the ros2 services from the webserver.
 - Once activated then press **L2+Up** to stand up the robot.
5. Robot sit down

- Once activated then press **L2+Down** to sit down the robot.

6. Single tap the **Rviz2 Launcher** on the screen to visualize the robot.

7.3 | Steamdeck - Termination

- Single tap the **ROS2 Clean** to terminate all ROS2 instances **only** in the steamdeck (It does not effect the robot).

7.4 | Steamdeck - Robot Controls

1. **L1** is the dead man's switch for slow speed.
2. **R1** is the dead man's switch for high speed.
3. **LJ** (Left Joystick) is to move in x and y directions.

Notice: Loss of connection or (laggy connection) may lead to the robot moving and/or not stopping immediately. Caution should be taken when using the Steamdeck for the operation of the robot!

7.5 | Steamdeck - Rviz2 Controls

1. **Right trackpad** is used as a mouse
2. **Left trackpad** can be moved up and down to zoom in and zoom out
3. **L2+R2** should be held when wanting to move the orbit in the xy plane.
4. **R2** should be held to orbit around the cursors origin

8 | Installation

8.1 | NVIDIA Orin NX Developer Kit (Inbuilt GO2 EDU)

Note: This repository should already be available and built on GO2's Nvidia board if it has been configured by the MYBOTSHOP team.

- Switch Go2 id with ctrl+h

```
go2_unit_27778
```

- Switch username

```
sudo hostnamectl set-hostname go2-unit-27778
```

- Update date and time, ensure you have internet in the robot. The quickest way is using the USB WiFi stick

```
sudo timedatectl set-timezone Europe/Berlin
sudo date -s "$(wget --method=HEAD -qSO- --max-redirect=0 \
google.com 2>&1 | sed -n 's/^ *Date: */p')"
```

- Do 'sudo apt-get update' if it fails usually removing the outdated packages resolves it. This is important

- Issues in latest go2's

```
sudo apt-key del F42ED6FBAB17C654
curl -sSL \
https://raw.githubusercontent.com/ros/rosdistro/master/ros.asc \
| sudo apt-key add -
sudo apt-get update
```

- Create directory for Go2 Workspace

```
sudo mkdir /opt/mybotshop
```

```
sudo chown -R unitree:unitree /opt/mybotshop
```

- Clone the repository, copy over to the Go2's PC and run the installer script

```
cd /opt/mybotshop/src/mybotshop/ && sudo chmod +x go2_install.sh
./go2_install.sh
cd /opt/mybotshop/src/mybotshop/go2_webserver
```

```
sudo chmod +x webserver_installer.sh
./webserver_installer.sh
```

■ Set VNC password

```
vncpasswd ~/.vnc/passwd
Password: mybotshop
Verify: mybotshop
Would you like to enter a view-only password (y/n)? y
Password: mybotshop
Verify: mybotshop
```

■ Build unitree sdk

```
cp -r /opt/mybotshop/src/third_party/unitree/04Aug2025_unitree_sdk2/ \
/opt/mybotshop

cd /opt/mybotshop/04Aug2025_unitree_sdk2/build

cmake .. && make && sudo make install
```

■ Build ros2 workspace

```
cd /opt/mybotshop && colcon build --symlink-install
source install/setup.bash
```

■ Change the .bashrc

```
# MYBOTSHOP
source ~/cyclonedds_ws/install/setup.bash
export RMW_IMPLEMENTATION=rmw_cyclonedds_cpp
export CYCLONEDDS_URI=~/.cyclonedds_ws/cyclonedds.xml
export PATH=/usr/local/cuda-11.4/bin:$PATH
export LD_LIBRARY_PATH=/usr/local/cuda-11.4/lib64:$LD_LIBRARY_PATH
#export LD_LIBRARY_PATH=/usr/local/lib:$LD_LIBRARY_PATH

source /opt/ros/foxy/setup.bash
source /opt/mybotshop/install/setup.bash
```

■ Run the startup installer

```
ros2 run go2_bringup startup_installer.py
```

■ Enable fastwebserver

```
cd /opt/mybotshop/src/mybotshop/go2_webserver/go2_webserver/  
rm -rf librosnode.py libaudiogen.py libwebserver.py \  
ignore_python_binartor.bash ignore_compile.py
```

■ Hesai attachment

- Goto the ip 192.168.123.20
- Set spin rate 300 and
- Set return mode strongest return
- Set azimuth 30 to 330

■ Openmanipulator

- change the udev serial to match
- ‘udevadm info -name=/dev/ttyUSB0 -attribute-walk’

■ Add D1 arm SDK

```
cp -r /opt/mybotshop/src/third_party/unitree/06Aug2025_d1_sdk/ \  
/opt/mybotshop  
  
cd /opt/mybotshop/06Aug2025_d1_sdk/build  
cmake .. && make && sudo make install
```

8.2 | Host PC

Note: foxy branch is required. Build go2_viz go2_description and other necessary packages, then

```
u20  
source /opt/ros/foxy/setup.bash  
source install/setup.bash  
export ROS_DOMAIN_ID=10  
export G02_NS=go2_unit_001  
unset RMW_IMPLEMENTATION  
ros2 launch go2_viz view_robot.launch.py
```

9 | Robotic Manipulator Safety Guidelines

When deploying robotic manipulators, it is imperative to prioritize safety procedures to mitigate risks and ensure secure operations. The following guidelines delineate key safety measures when working with robotic manipulators:

9.1 | Work Area Safety

- Maintain a clean and well-organized work area. Cluttered or inadequately lit spaces can impede the proper functioning of sensors and hinder precise manipulation.
- Avoid operating robotic manipulators in hazardous environments, such as those containing corrosive substances, extreme temperatures, or sharp objects that may damage the manipulator.
- Ensure that only authorized personnel are present in the vicinity during manipulator operation to prevent interference and ensure a safe working environment.

9.2 | Electrical Safety

- Ensure the manipulator's power system adheres to electrical safety standards. Regularly inspect and maintain power components to prevent malfunctions.
- Implement safeguards to protect the manipulator from adverse environmental conditions, such as exposure to moisture or extreme temperatures.
- Regularly inspect power cables and connections, promptly replacing damaged components to minimize the risk of electrical issues.

9.3 | Manipulation Safety

- Implement collision detection systems to prevent unintended contact with objects, humans, or other equipment during manipulation tasks.
- Define and enforce safety zones around the manipulator's workspace to minimize the risk of unintended interactions with personnel or other objects.
- Regularly calibrate and test the manipulator's sensors and systems to ensure precise and reliable performance during manipulation tasks.

9.4 | Emergency Response

- Install an emergency stop mechanism to swiftly halt manipulator operation in unforeseen circumstances or emergencies.
- Clearly mark and communicate emergency stop locations within the manipulator's operational area.

- Conduct regular emergency response drills to ensure personnel are familiar with procedures for handling unexpected situations during manipulator operation.

9.5 | Data Security and Privacy

- Implement robust cybersecurity measures to safeguard the manipulator's control systems and data from unauthorized access or manipulation.
- Ensure compliance with privacy regulations when collecting, storing, or transmitting data captured by the manipulator's sensors.

9.6 | Human Interaction Safety

- Integrate sensors and communication systems to detect and respond to the presence of humans in the manipulator's vicinity.
- Clearly communicate the manipulator's operational status and intentions using visual and audible signals to alert nearby individuals.
- Establish protocols for safe human-robot collaboration, particularly in shared workspaces where manipulators are in operation.

9.7 | Residual Risks

Despite the implementation of safety measures, certain residual risks may persist. These include:

- Impairment of sensor functionality.
- Risk of collisions during complex manipulation tasks.
- Cybersecurity vulnerabilities.
- Unintended human interactions due to unforeseen circumstances.

Robotic manipulators are sophisticated technologies that demand correct usage to avoid accidents and ensure a secure environment. Please adhere to proper procedures diligently, prioritizing both precision and safety.

10 | Autonomous Mobile Robot Safety Guidelines

When deploying autonomous mobile robots, prioritizing safety procedures is essential to prevent accidents and ensure secure operations. The following guidelines outline key safety measures when working with an autonomous mobile robot:

10.1 | Work Area Safety

- Maintain a clean and well-lit work area. Cluttered or poorly illuminated spaces can impede the proper functioning of sensors and navigation systems.
- Avoid operating autonomous mobile robots in explosive atmospheres, such as areas with flammable liquids, gases, or dust. The robot's components may pose a risk in such environments.
- Keep bystanders and unauthorized personnel at a safe distance during robot operation to prevent interference with autonomous navigation.

10.2 | Electrical Safety

- Ensure the robot's power system adheres to electrical safety standards. Regularly inspect and maintain power components to prevent malfunctions.
- Implement mechanisms to protect the robot from adverse weather conditions, such as rain or wet environments.
- Regularly inspect power cables and connections and replace damaged components promptly to minimize the risk of electrical issues.

10.3 | Navigation Safety

- Implement obstacle detection and avoidance systems to prevent collisions with objects, people, or other robots.
- Define and enforce safety zones within the robot's operational area to minimize the risk of unintended interactions with personnel or other equipment.
- Regularly calibrate and test the robot's navigation sensors and systems to ensure accurate and reliable performance.

10.4 | Emergency Response

- Install an emergency stop mechanism to quickly halt the robot's operation in case of unforeseen circumstances or emergencies.
- Clearly mark and communicate emergency stop locations throughout the robot's operational area.

- Conduct regular emergency response drills to ensure personnel are familiar with procedures for handling unexpected situations.

10.5 | Data Security and Privacy

- Implement robust cybersecurity measures to protect the robot's control systems and data from unauthorized access or manipulation.
- Ensure compliance with privacy regulations when collecting, storing, or transmitting data captured by the robot's sensors.

10.6 | Human Interaction Safety

- Integrate sensors and communication systems to detect and respond to the presence of humans in the robot's vicinity.
- Clearly communicate the robot's operational status and intentions using visual and audible signals to alert nearby individuals.
- Establish protocols for safe human-robot collaboration, especially in shared workspaces.

10.7 | Residual Risks

Despite the implementation of safety measures, certain residual risks may persist. These include:

- Impairment of sensor functionality.
- Risk of collisions in crowded or dynamic environments.
- Cybersecurity vulnerabilities.
- Unintended human interactions due to unforeseen circumstances.

Autonomous Mobile Robots (AMR) are advanced technologies that require correct usage to avoid accidents and ensure a secure environment. Learn and follow the proper procedures diligently; prioritize both quality and safety.

11 | Support

MYBOTSHOP



QUADRUPED



In case of any issues, help can be asked in:

1. **MYBOTSHOP FORUMS** (forum.mybotshop.de)
2. **MYBOTSHOP SUPPORT** (support@mybotshop.de)

Further information on different robots can be viewed in:

1. **MYBOTSHOP DOCS** (docs.mybotshop.de)
2. **QUADRUPED DOCS** (docs.quadruped.de)

12 | Tutorials

A collection of open-source training material to familiarize oneself with the ROS2 environment.

12.1 | Ubuntu Tutorials

1. [Ubuntu Documentation](#) - Official Ubuntu Documentation.

- help.ubuntu.com/

2. [Ubuntu ROS2 Documentation](#) - Official Ubuntu ROS2 Documentation.

- ubuntu.com/tutorials/getting-started-with-ros-2#1-overview

12.2 | ROS2 — Tutorials

1. [ROS2 Tutorial List](#) - ROS2 Tutorial List.

- docs.ros.org/en/foxy/Tutorials.html

2. [ROS2 Basics Training](#) - ROS2 Basics Training.

- industrial-training-master.readthedocs.io/en/melodic/_source/session7/ROS2-Basics.html

3. [ROS2 foxy Wiki](#) - Official ROS2 documentation and tutorials.

- docs.ros.org/en/foxy/index.html

4. [ROS Navigation2 Wiki](#) - Official ROS navigation documentation and tutorials.

- navigation.ros.org/

5. [ROS2 Manipulation Wiki](#) - Official ROS2 manipulation documentation and tutorials.

- moveit.picknik.ai/main/index.html

6. [ROS2 Discourse](#) - Official ROS2 discussion platform.

- discourse.ros.org/c/ng-ros/25

7. [ROS2 Courses](#) - Official ROS2 courses.

12.3 | ROS2 — Additional Resources

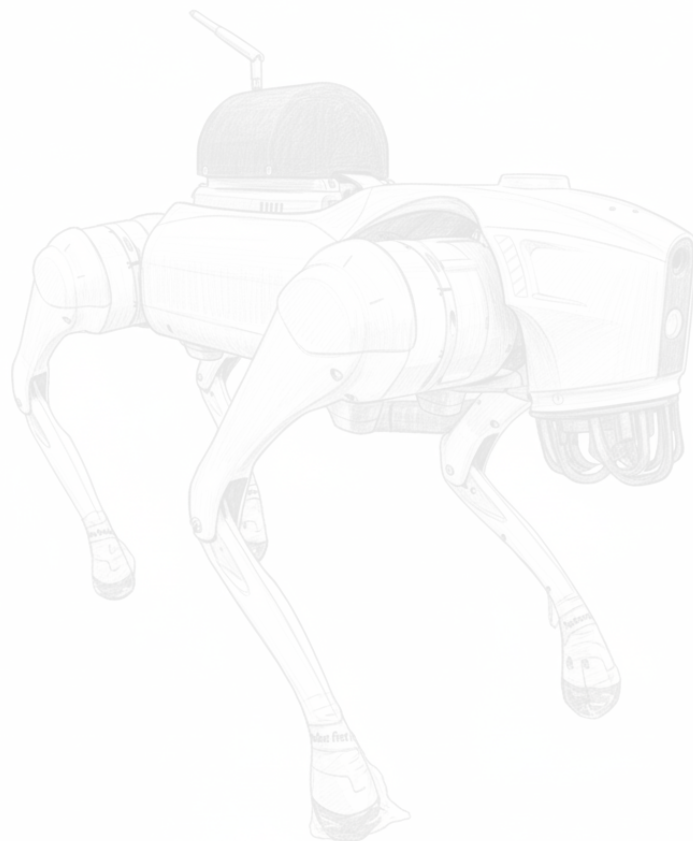
1. [ROS2 Installation Guide](#) - Documentation.

2. [ROS2 Pertinent Repositories](#) - GitHub Repository.

3. [ROS2 Concepts](#) - Documentation.

4. [ROS2 How to guides](#) - Documentation.

5. [ROS2 for Beginners - The Construct](#) - YouTube playlist.
6. [ROS2 foxy Crash Course](#) - YouTube playlist.
7. [ROS2 Basic Learning](#) - YouTube playlist.
8. [ROS Tutorials - ETH Zurich](#) - YouTube playlist. ²
9. [ROS Workshop](#) - YouTube Video.



²It is in ROS1, however, provides basic understanding of the ROS paradigm.

13 | ROS — MYBOTSHOP/QUADRUPED GmbH Trainings

MYBOTSHOP/QUADRUPED GmbH offers paid training as well for basic ROS training as well as for transitional training. An example of the training schedule is provided below.

13.1 | ROS1 to ROS2 δ

The provided course would be a general concise transitional course from ROS1 to **ROS2 foxy**. The training would be geared towards basic practical implementation and deployment of ROS2 foxy with those who are **well** familiar with ROS1. The average time for the training is 2—3 days with a focus more on practicality and less on theory.

ROS1 to ROS2 δ - Pre-requisities

- ROS1 (Kinetic/Melodic/Noetic)
 - Working experience and concepts
 - Control concepts
 - Navigation stack working
 - URDF
 - Transforms (TFs)
 - Nodes — Nodelets
 - Topics — Parameters — Services — Msgs — Action Service
- Ubuntu 22.04 installed
- ROS2 Humble Installed
- Internet Connection (Package installation)

13.1.1 | ROS1 to ROS2 δ - List of Deliverables

1. Presentation slides
2. Simulation-based tasks
3. ROS2 Template Packages

ROS1 to ROS2 δ — Basics — Day 1

- Introduction to ROS2 Humble
 - Reason of ROS2 (T)
 - Key differences (T)

- ☐ New architecture (T)
- ☐ Why DDS? and its Robustness and longevity! (T)
- ☐ ROS2 Control (Brief Overview) (T)
- ☐ ROS2 Package Build (T)
- ☐ ROS Bridge (T)
- ☐ ROS2 Simulation (P)
- ☐ ROS2 Create Package (P)

■ ROS2 Nodes — Lifecycle nodes — Components — Topics

- ☐ ROS2 nodes (T)
- ☐ ROS2 launch files (T)
- ☐ Create ROS2 Node (P)
- ☐ Create ROS2 Launch file (P)
- ☐ ROS2 Sensor Fusion (P)

■ ROS2 Parameters — Msgs — Services — Action-server

- ☐ Item Differences (T)
- ☐ ROS2 Parameters (T)
- ☐ Create ROS2 Parameter Loading (P)

ROS1 to ROS2 δ — Autonomous Navigation — Day 2

■ ROS2 Navstack

- ☐ Differences (T)
- ☐ Setup (P)
- ☐ Mapping (P)
- ☐ Localization (P)
- ☐ Planner Tuning (P)
- ☐ Custom Robot Integration (P)
- ☐ Debugging (P)
- ☐ Tools Usage (P)

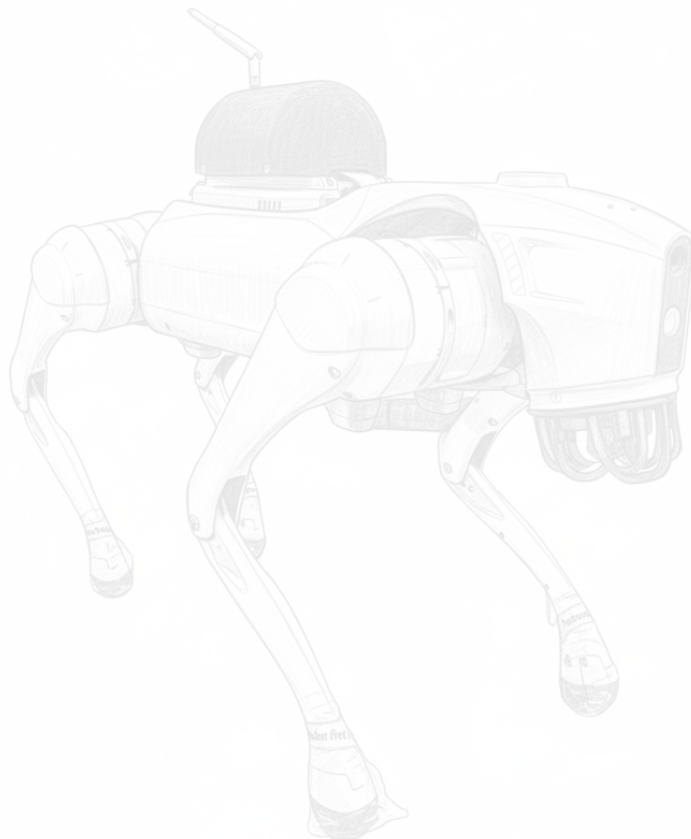
■ Personal Experience

- ☐ Issues faced
- ☐ Debugging
- ☐ Tools Usage

13.2 | Additional Trainings

In addition to the ROS1 to ROS2 δ trainings, we have trainings for:

1. Ubuntu 20 — ROS1 Noetic (1—2 days)
2. Ubuntu 22 — ROS2 Humble (1—2 days)
3. Robot operational training (Specialized Software Solution Projects) (1—2 days)



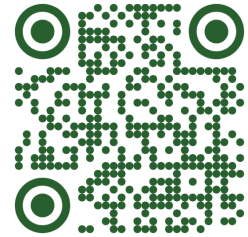
Resources and Contact Information

For your convenience, we have compiled the most important resources related to the robot platform. Each link can be accessed directly or by scanning the corresponding QR code.

Documentation

Our official documentation contains detailed setup instructions, usage guidelines, and troubleshooting information. Access the latest version here:

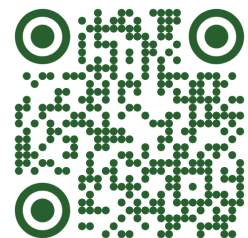
<https://www.docs.quadruped.de/index.html>



Forum

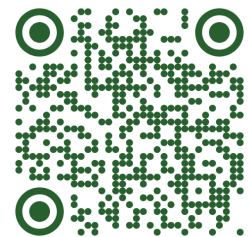
Connect with other developers and researchers in our community forum.

Here you can ask questions, share experiences, and stay up to date: <https://forum.mybotshop.de/>



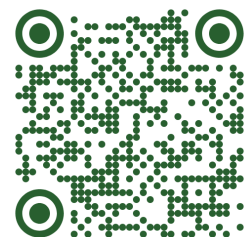
Rentals

If you would like to test or temporarily use a robot, we also offer rental services. Browse our available systems here: https://www.mybotshop.de/Vermietung_3



Newsletter

Stay informed about product updates, events, and special offers by subscribing to our newsletter: https://www.mybotshop.de/Newsletter_1



Contact Information

■  Vertrieb / Sales: +49 (0) 2271 588 93 10 —  sales@mybotshop.de

■  Support / After Sales: +49 (0) 2271 588 93 20 —  support@mybotshop.de

■  Buchhaltung / Accounting: +49 (0) 2271 588 93 80 —  accounting@mybotshop.de

■  Allgemeine Anfragen / General inquiries: +49 (0) 2271 588 93 0 —  info@mybotshop.de

QUADRUPED ROBOTICS

RMA SUPPORT

How to Raise a Return Merchandise Authorization (RMA) Case

At QUADRUPED Robotics, we are committed to providing excellent support to resolve your issues promptly. If you encounter any problems with our products, please follow the steps below to raise a Return Merchandise Authorization (RMA) case:

Step 1: Open a Topic in Our Forum

1. Visit [MYBOTSHOP Forum](#)
2. Open a new topic describing your issue in detail. Our support team and community members will attempt to assist you in resolving the problem remotely.

Step 2: Contact Support for RMA Number

If the issue cannot be resolved remotely through our forum:

1. Email us at support@quadruped.de
2. Provide a detailed description of the issue and include any troubleshooting steps already taken.
3. Our support team will evaluate your case and, if necessary, issue an RMA number along with a return address or a shipping label.

Important Shipping Information

Please note:

- All goods must be sent back to the following address:
 - **QUADRUPED Robotics GmbH**
 - **c/o MYBOTSHOP GmbH**
 - **Willy-Messerschmitt-Strasse 1250126**
 - **Bergheim Germany**
- Do not ship goods without obtaining an RMA number. Parcels sent without an RMA number will be declined and returned to the sender.
- Do **not ship goods** to QUADRUPED Robotics GmbH's Office in Leverkusen!

Help Resources

In case of any issues, help can be taken from any of the following forums/resources:

- [MYBOTSHOP Forum](#): The forum is actively monitored by support teams at [MYBOTSHOP GmbH](#).
- [Github Issues](#): Issues can be reported in the repository.
- Online QA Sites: Questions can be asked on any of the QA sites, like [StackOverflow](#) and [Answers ROS](#).